

IPv6: The New Address

Until IPv6 completely supplants IPv4, a number of transition mechanisms are needed to enable IPv6-only hosts to reach IPv4 services and to allow isolated IPv6 hosts and networks to reach the IPv6 Internet over the IPv4 infrastructure

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Every computer or other device that connects to the Internet requires a unique Internet protocol (IP) address. Internet protocol version 4 (IPv4) uses 32-bit addresses that allow 2^{32} or about 4.3 billion unique IP addresses. The IP address space is managed by the Internet Assigned Numbers Authority (IANA) globally, and by five Regional Internet Registries (RIRs) responsible in their designated territories for assignment to end users and local Internet registries, such as

Internet service providers—according to Wikipedia.

In the late 1970s, when the IPv4 address space was designed, it was believed that its 4.29 billion addresses would be sufficient to cover any future growth of the Internet. However, with advances in technology and an allocation practice that did not anticipate the explosion of hosts on the Internet, by 1992 it was clear that a replacement would be necessary.

IPv6: the saviour

In December 1998, the Internet Engineering Task Force (IETF) developed IPv6 to deal with exhaustion of IPv4 addresses, and described it in Request For Comments (RFC 2460). IPv6 uses an addressing scheme

foreseeable future.

Addressing. Use of 128 bits allows multiple levels of hierarchy and flexibility in designing hierarchical addressing and routing, which the current IPv4 Internet lacks. IPv6 addresses are represented in colon-hexadecimal format. The addresses are divided along 16-bit boundaries. The 16-bit blocks are converted into 4-digit hexadecimal numbers and separated by colons.

IPv6 addresses are classified into several groups such as unicast (one-to-one), multicast (one-to-many) and anycast (one-to-nearest). IPv6 does not define a broadcast address and all types of IPv4 broadcast addresses are performed in IPv6 using multicast addresses.

Packet. An IPv6 packet contains IPv6 header, extension headers and upper-layer protocol data unit. The IPv6 header is always present and fixed at 40 bytes in size.

IPv4 and IPv6 interoperability and transition

According to Wikipedia, until IPv6 completely supplants IPv4, a number of transition mechanisms are needed to enable IPv6-only hosts to reach IPv4 services and to allow isolated IPv6 hosts and networks to reach the IPv6 Internet over the IPv4 infrastructure. A wide range of techniques have been identified and implemented by IETF for this purpose. The techniques can be broadly categorised as:

1. Dual IP stack techniques that allow IPv4 and

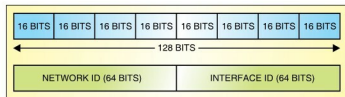


Fig. 1: Addressing scheme in IPv6

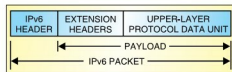


Fig. 2: Structure of an IPv6 packet

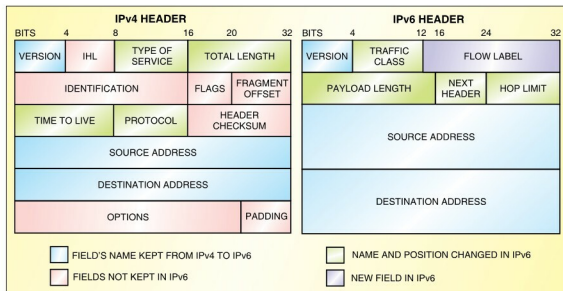


Fig. 3: IPv4 and IPv6 headers